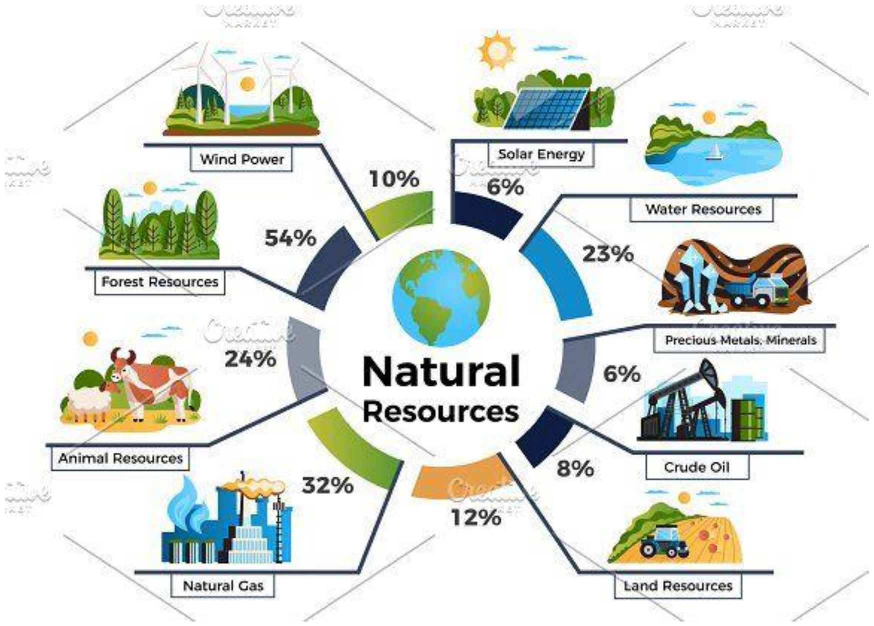
Conservation of Mineral Resources and Loss of Mineral in Mining

B S Choudhary

Department of Mining Engineering

IIT(ISM) Dhanbad



Contents

- Means of conservation and limitations in the scope of Conservation
- Classification and incorporation of losses
- Coefficient of completeness of mineral extraction
- Dilution and recovery.

MCDR-2017:MCDR2017.pdf

Means of conservation and limitations in the scope of Conservation

Means of conservation

- Use of minerals in a planned and sustainable manner, recycling of metals
- Mineral resources should be conserved because the geological process of mineral formation is quite slow due to which the rate of replenishment is infinitely small whereas the rate of consumption is quite high. Mineral resources found on the earth surface are limited in number and are exhaustible
- Conservation of mineral resources includes improved efficiency, substitution, and the 3 Rs of sustainability, reduce, reuse, and recycle. Improved efficiency applies to all features of mineral use including mining, processing, and creation of mineral products.

Scope of Conservation

1) Techniques to utilize lower-grade mineral deposits, often without increase in real costs;
2) Development of substitutes for "scarce" minerals; and
3) More efficient and effective use of the available primary and secondary (scrap) supplies of the minerals

- Mineral markets and in turn prices are influenced primarily by unexpected changes in mineral demand
- Understanding Importance in Use or the Impact of a Supply Restriction: **The greater the difficulty, expense, or time it takes for material substitution to occur, the more critical a mineral is to a specific application, the greater is the impact of a supply restriction.**
- Lost production due to lack of availability or higher costs (use will be concentrated in higher-valued uses of a mineral or mineral product);
- Higher costs of production, which producers may or may not be able to pass along to consumers;
- Slower growth than otherwise in emerging-use industries;
- Less employment than otherwise in industries using minerals and mineral products as inputs;

Unwise uses

- Destructive use -- use of resources should be in a proper order of priority (e.g. use of renewable resources when possible, instead of exhaustible minerals; use of plentiful resources rather than scarce ones).
- Underuse of renewable resources -- practice the full utilization of their capacity to produce and still maintain their regenerative capacity.
- Mismanagement of non-renewable resources -- full recovery of all minerals in a given deposit is a better policy than selective mining of high-grade ore.
- Wrong use of extractive and final products -- get the maximum level of service from the extractive products (e.g. reinjection of natural gas back to the reservoir, rather than flaring it).

Methods of conservation

- Continuous search for new sources of raw materials
- Choice of correct mining method
- Utilization of unmarketable, by ore dressing
- Use of re-use of scrap
- Recovery of all associated elements
- Economy in the use of mineral
- Finding new substitutes
- Newer methods of mining and utilize thinner ore
- Prohibiting non-essential use
- New ideas for conservation and promotion

- Stowing
- Coal washeries
- Efficient production/ clean coal production
- Recovery of by products
- Blending

Sustainable Development Principles

Economic Sphere

- Maximize human well-being.
- Ensure efficient use of all resources, natural and otherwise, by maximizing rents.
- Seek to identify and internalize environmental and social costs.
- Maintain and enhance the conditions for viable enterprise.

Social Sphere

- Ensure a fair distribution of the costs and benefits of development for all those alive today.
- Respect and reinforce the fundamental rights of human beings, including civil and political liberties, cultural autonomy, social and economic freedoms, and personal security.
- Seek to sustain improvements over time; ensure that depletion of natural resources will not deprive future generations through replacement with other forms of capital.

Environmental Sphere

- Promote responsible stewardship of natural resources and the environment, including remediation of past damage.
- Minimize waste and environmental damage along the whole of the supply chain.
- Exercise prudence where impacts are unknown or uncertain.
- Operate within ecological limits and protect critical natural capital.

Governance Sphere

- Support representative democracy, including participatory decision-making.
- Encourage free enterprise within a system of clear and fair rules and incentives.
- Avoid excessive concentration of power through appropriate checks and balances.
- Ensure transparency through providing all stakeholders with access to relevant and accurate information.
- Ensure accountability for decisions and actions, which are based on comprehensive and reliable analysis.
- Encourage cooperation in order to build trust and shared goals and values.
- Ensure that decisions are made at the appropriate level, adhering to the principle of subsidiarity where possible.

Key Challenges in mining

Viability of the Minerals Industry. The minerals industry cannot contribute to sustainable development if companies cannot survive and succeed. This requires a safe, healthy, educated, and committed work force; access to capital; a social licence to operate; the ability to attract and maintain good managerial talent; and the opportunity for a return on investment.

The Control, Use, and Management of Land. Mineral development is one of a number of often competing land uses. There is frequently a lack of planning or other frameworks to balance and manage possible uses. As a result, there are often problems and disagreement around issues such as compensation, resettlement, land claims of indigenous peoples, and protected areas.

Minerals and Economic Development. Minerals have the potential to contribute to poverty alleviation and broader economic development at the national level. Countries have realized this with mixed success. For this to be achieved, appropriate frameworks for the creation and management of mineral wealth must be in place. Additional challenges include corruption and determining the balance between local and national benefits.

Local Communities and Mines. Minerals development can also bring benefits at the local level. Recent trends towards, for example, smaller work forces and outsourcing affect communities adversely, however. The social upheaval and inequitable distribution of benefits and costs within communities can also create social tension. Ensuring that improved health and education or economic activity will endure after mines close requires a level of planning that has too often not been achieved.

Mining, Minerals, and the Environment. Minerals activities have a significant environmental impact. Managing these impacts more effectively requires dealing with unresolved issues of handling immense quantities of waste, developing ways of internalizing the costs of acid drainage, improving both impact assessment and environmental management systems, and doing effective planning for mine closure.

An Integrated Approach to Using Minerals. The use of minerals is essential for modern living. Yet current patterns of use face a growing number of challenges, ranging from concerns about efficiency and waste minimization to the risks associated with the use of certain minerals. Companies at different stages in the minerals chain can benefit from learning to work together

Access to Information. Access to information is key to building greater trust and cooperation. The quality of information and its use, production, flow, accessibility, and credibility affect the interaction of all actors in the sector. Effective public participation in decision-making requires information to be publicly available in an accessible form.

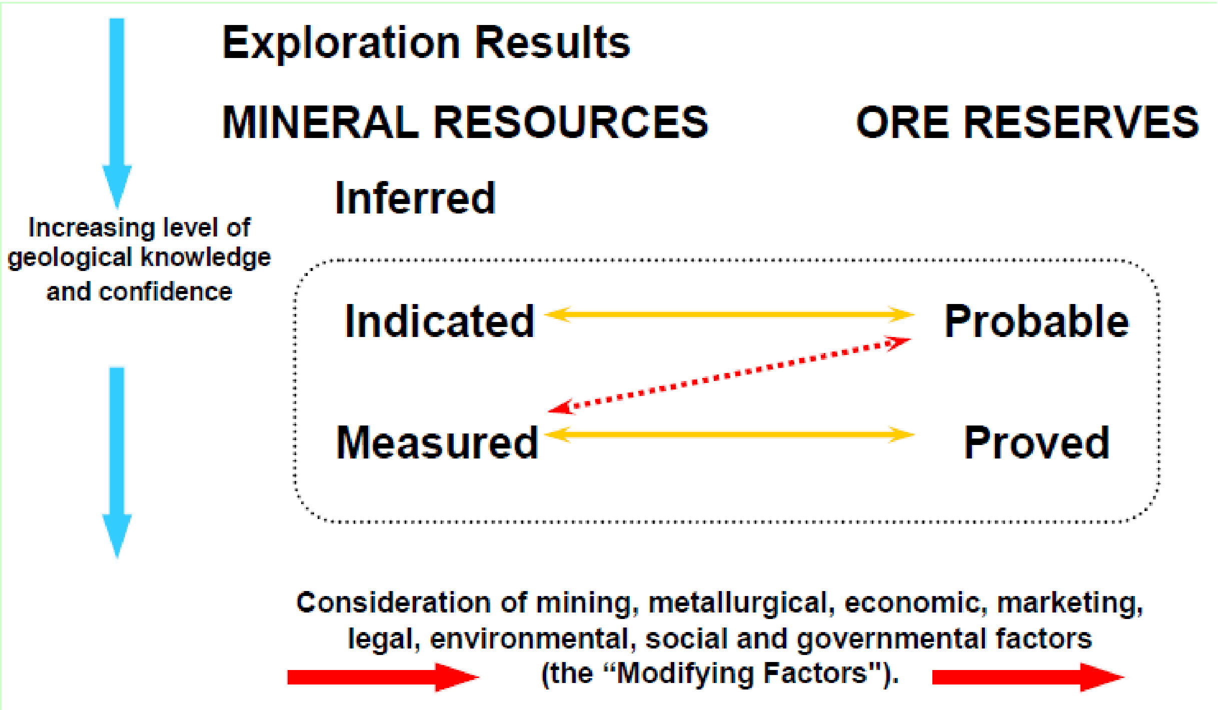
Artisanal and Small-Scale Mining. Many millions of people make their living through artisanal and small-scale mining (ASM). It often provides an important, and sometimes the only, source of income. This part of the sector is characterized by low incomes, unsafe working conditions, serious environmental impacts, exposure to hazardous materials such as mercury vapours, and conflict with larger companies and governments.

Sector Governance: Roles, Responsibilities, and Instruments for Change. Sustainable development requires new integrated systems of governance. Most countries still lack the framework for turning minerals investment into sustainable development: these need to be developed. Voluntary codes and guidelines, stakeholder processes, and other systems for promoting better practice in areas where government is unable to exercise an effective role as regulator are gaining favour as an expedient to address these problems. Lenders and other financial institutions can play a pivotal role in driving better practice.

Coefficient of completeness of mineral extraction

- The extraction ratio is defined as the volume of ore/coal mined divided by the total volume of ore/coal within a reserve

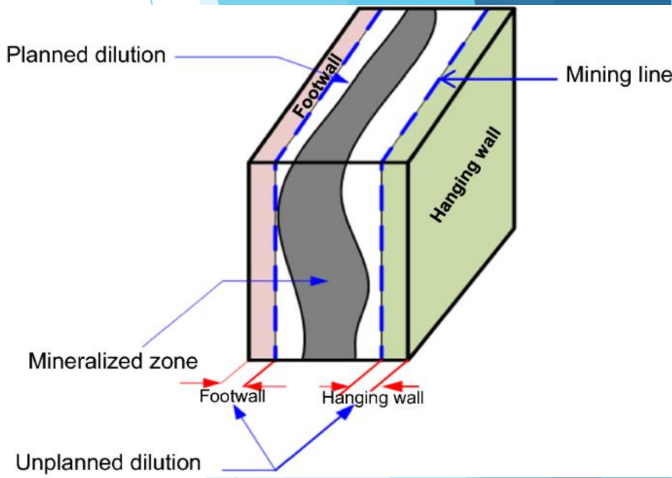
The location and shape of the deposit, strength of the rock, ore grade, mining costs, and current market price of the commodity are some of the determining factors for selecting which mining method to use.



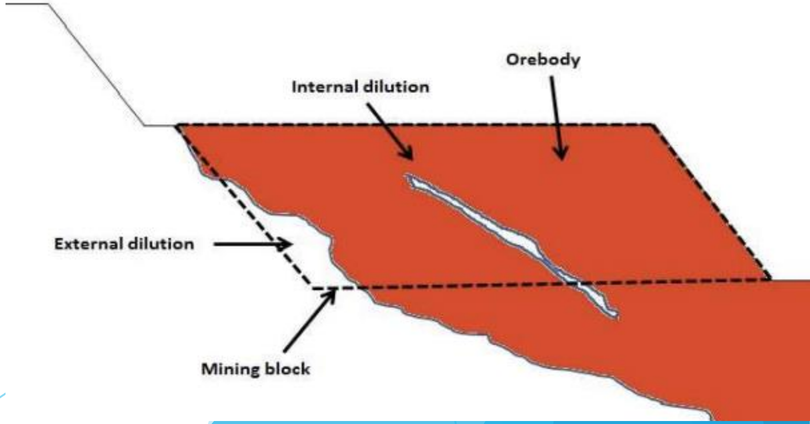
Dilution and Ore recovery

- Dilution refers to the waste material that is not separated from the ore during the operation and is mined with ore. This waste material is mixed with ore and sent to the processing plant.
- Dilution is the result of mixing non-ore grade material with ore-grade material during production, generally leading to an increase in tonnage and a decrease in mean grade relative to original expectations.
- Dilution can be defined as the ratio of the tonnage of waste mined and sent to the mill for processing over the combined the total tonnage of ore and waste that are milled.

$$Dilution(\%) = \frac{TonnesWasteMilled}{TonnesOreMilled + TonnesWasteMilled} * 100\%$$



Planned and Unplanned Dilution



Internal and External Dilution

Factors of Dilution

- Mine dilution occurs due to the mining method selected and from overbreak during the mining process. There are multiple considerations in terms of dilution. Mining methods such as block caving, sublevel stoping and room-and-pillar have mining methods which are more predictable, where dilution can be modeled using empirically generated equations.
- The major factors which have a direct effect on dilution are as follows:
  - Mine Depth: Methods with greater selectivity exhibit lower dilution
  - Rock Competency : More competent rock will be less susceptible to sloughing and overbreaking
  - Ore Type : Defines the selective and effective dilution parameters
  - Ground Support: Support can be used to maintain ore and waste surfaces, limiting the amount of dilution

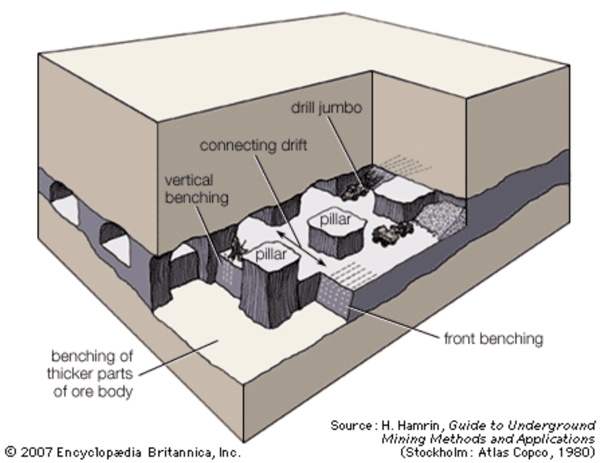
  Self-supported openings are more selective and have lower dilutions than block caving with typical dilution ranges from 5% to 15%. Additional factors which influence dilution to a lesser extent are as follows:

  - Rock Mechanics: Mechanical parameters and technical ability lead to increased dilution to account for
  - Ore Geometry : Layout of the ore in skewed orientation leads to increases in unplanned dilution
  - Hanging-wall Dip : The likelihood of wall slabbing and release of wedges will depend on wall dip relative to the orientation of lamination and joints
  - Geotechnical: Parameters are increased lead to an increase in dilution values
  - Stope Span: Larger stop

ORE RECOVERY

- Ore recovery is based on the material within the model that is left behind to provide structural support, thereby not being recovered.
- More specifically, ore recovery can be defined by the percentage of minable reserves extracted in the mining process. The issue of balancing dilution and ore recovery is a challenging one as profitability is to be optimized while not effecting the efficiency of operation. It has been noted that instead of utilizing labour intensive mining methods to high tonnage bulk mining has decreased the ability to control ore recovery.
- In a practical underground and open pit setting, the ore recovery is affected by three main factors as follows:
    - Ore wedges located at the top and bottom sills of the panel are unable to be extracted as they provide access to mined-out areas in backfill and provide support. This is comparable to room and pillar operations in which pillars are also left for support.
    - Ore left against backfill, as there exists a certain amount of ore adjacent to the backfill that is unable to be extracted due to irregularities of the ore-backfill contact surface
    - Oxidization of ore stuck to the footwall is a direct consequence of the extraction sequence. In this, ore blasted at the beginning of the extraction phase must stay within the stope until the last slice of ore is extracted.

$$Recovery(\%) = \frac{ExtractedOre}{ExtractedOre + UnextractedOre} * 100\%$$



Thanks